

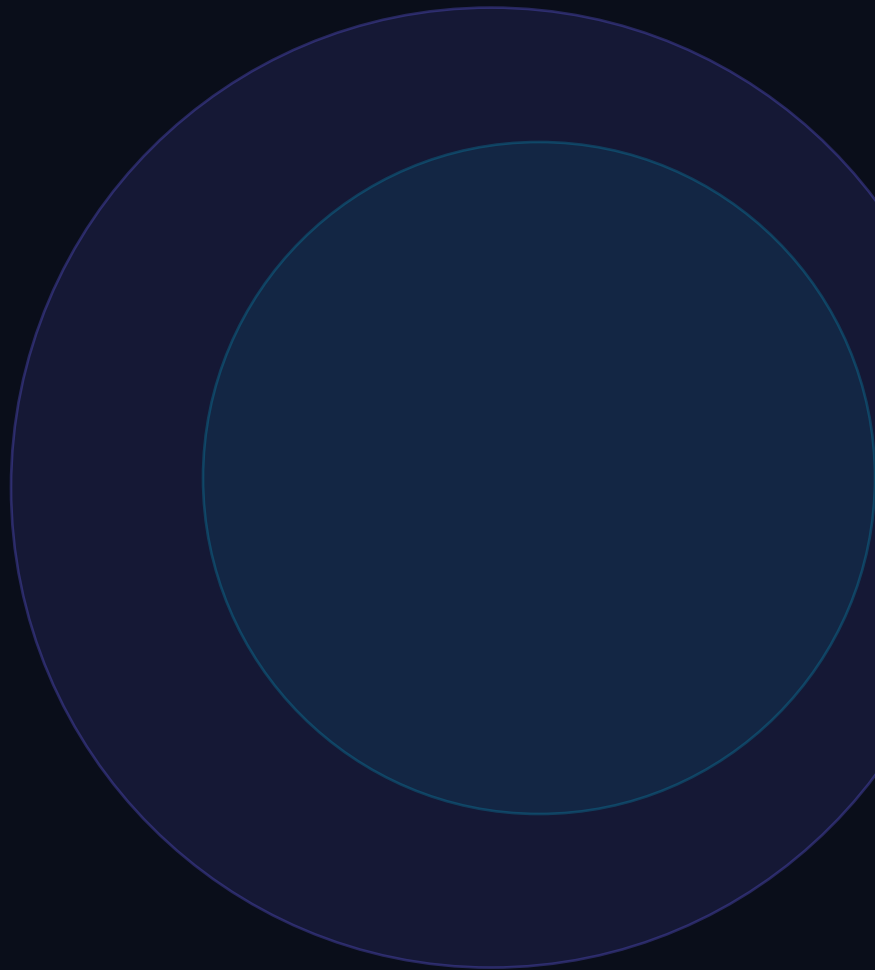


P R O J E C T

LUMINAR

The Augmented Reality Data Projector

Visualizing the Unseen. Directly on the World Around Us.



Data is Trapped on Screens

Critical data is isolated

Thermal, structural, and logistical data confined to small, disconnected screens.

Cognitive gap

Technicians constantly look away from their work—creating errors and safety risks.

Collaboration barrier

Teams huddle around tiny screens, unable to share a common data view.

AR headsets fall short

Expensive, cumbersome, isolating, and impractical in rugged field environments.



Data Liberated



Break data free from the screen

Data belongs in the world, not behind glass.



Shared spatial awareness

A team sees, shares, and interacts with the same overlay—simultaneously.



Intuitive & collaborative by design

Context-aware data that makes sense right where the work happens.



"What if a team could see, share, and interact with the same data overlay — in the same physical space — at the same time?"

The LUMINAR Promise

THE PRODUCT

Introducing Project LUMINAR

A portable, handheld device that Scans, Analyzes, and Projects data directly onto any surface.



SCAN

LiDAR · Infrared · RGB Camera

Captures geometry and properties of the environment with precision sensor arrays.



ANALYZE

Onboard System on a Chip

Powerful SoC processes all sensor data in real-time with dedicated NPU acceleration.



PROJECT

Laser-based Pico Projector

High-definition overlay with perfect geometric alignment onto any surface.

The Technology Stack



HARDWARE CORE



Laser Pico Projector

High brightness, focus-free beam



ToF / LiDAR Depth Sensor

Precise spatial mapping



Modular Sensor Bay

IR/Thermal camera (swappable)



ARM SoC with GPU & NPU

Real-time processing power



Swappable Battery

Field-replaceable high-density pack



SOFTWARE CORE (CS ENGINE)



Real-time OS & Firmware

Lag-free hardware communication



Sensor Fusion Engine

Combines LiDAR, IR, and RGB streams



SLAM Algorithm

3D perception & device positioning



Projection Mapping Module

Warp image to uneven surfaces in real-time



Data Visualization API

Renders compelling 'reinfographics'

Functional Requirements

What the system MUST DO — observable, testable behaviours expected by users and stakeholders.

FR-01

Real-Time Sensor Capture

The system shall capture thermal, depth (LiDAR/ToF), and RGB data simultaneously at ≥ 30 fps with a maximum end-to-end pipeline latency of 100 ms.

FR-02

Surface Projection Mapping

The system shall geometrically warp the projected output to conform to arbitrary non-planar surfaces in real-time, using depth data for keystone and distortion correction.

FR-03

Thermal Visualisation Overlay

The system shall render a colour-coded thermal heat map (e.g. IRONBOW palette) aligned pixel-accurately to the physical object, with configurable temperature thresholds and alert colours.

FR-04

Sensor Fusion & SLAM

The system shall fuse LiDAR, RGB, and IMU data streams using a Sensor Fusion Engine to maintain real-time 6-DoF pose estimation of the device relative to the scene.

FR-05

Modular Data Visualisation API

The system shall expose a documented REST/gRPC API allowing third-party developers to register custom sensor modules and define new 'reinfographic' rendering pipelines.

FR-06

Remote Database Integration

The system shall retrieve asset metadata (part numbers, maintenance history, schematics) from a central database via Wi-Fi/LTE within ≤ 2 seconds upon object recognition.

Non-Functional Requirements

Quality attributes that govern HOW the system performs — the engineering constraints and standards it must meet.

Performance

NFR-01

- Scan-to-projection pipeline latency: ≤ 100 ms end-to-end
- Projection frame rate: ≥ 30 fps sustained; ≥ 60 fps target
- Object recognition response: ≤ 500 ms from trigger

Reliability & Availability

NFR-02

- MTBF (Mean Time Between Failures): ≥ 500 operating hours
- Graceful degradation: device must continue projecting if DB connection fails
- Sensor fault detection and user alert within 200 ms

Portability & Durability

NFR-03

- IP54 rated enclosure: dust and splash resistant for field use
- Operating temperature range: -10°C to $+55^{\circ}\text{C}$
- Drop resistance: survive 1.2 m drop onto concrete (MIL-STD-810)

Usability

NFR-04

- A new user must reach operational proficiency within ≤ 15 minutes
- UI must conform to WCAG 2.1 AA contrast ratio standards
- Single-handed operable interface — no two-hand interaction required

Security

NFR-05

- All data transmissions encrypted via TLS 1.3 or higher
- Firmware updates must be cryptographically signed (RSA-2048)
- Local sensor data must not persist beyond the active session by default

Maintainability & Scalability

NFR-06

- Codebase must achieve $\geq 80\%$ unit test coverage (CS Engine modules)
- New sensor modules must be integrable without core firmware changes
- API versioning enforced; v1 endpoints supported for ≥ 24 months after v2 release

Industrial Maintenance & Safety

01

Point & Scan

Technician points LUMINAR at a complex engine or industrial system.

02

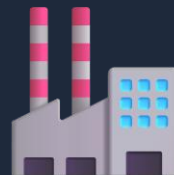
Live Thermal Projection

Device projects a live thermal heat map—overheating components glow bright red instantly.

03

Schematic Overlay

One button press switches to part numbers, maintenance history, and service records.



● LIVE THERMAL OVERLAY ACTIVE

Component #7B | TEMP: 247°C ⚠ CRITICAL

Part: TRB-2201-A | Last service: 47 days ago

📄 Schematic overlay with maintenance history fetched from central database automatically.

Emergency Services

01

Scan the Wall

Firefighter scans a wall in a smoke-filled room with LUMINAR.

02

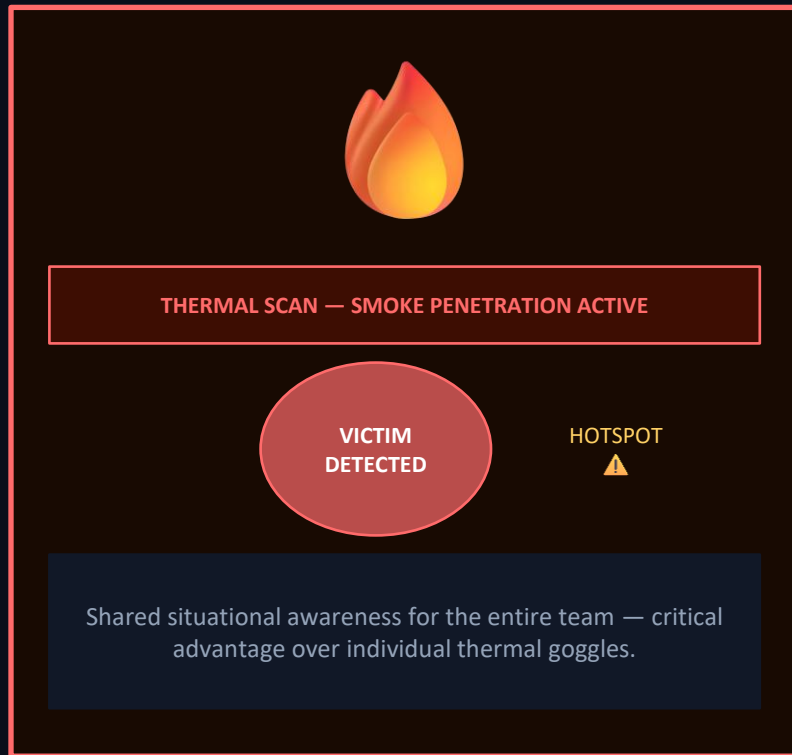
See Through Smoke

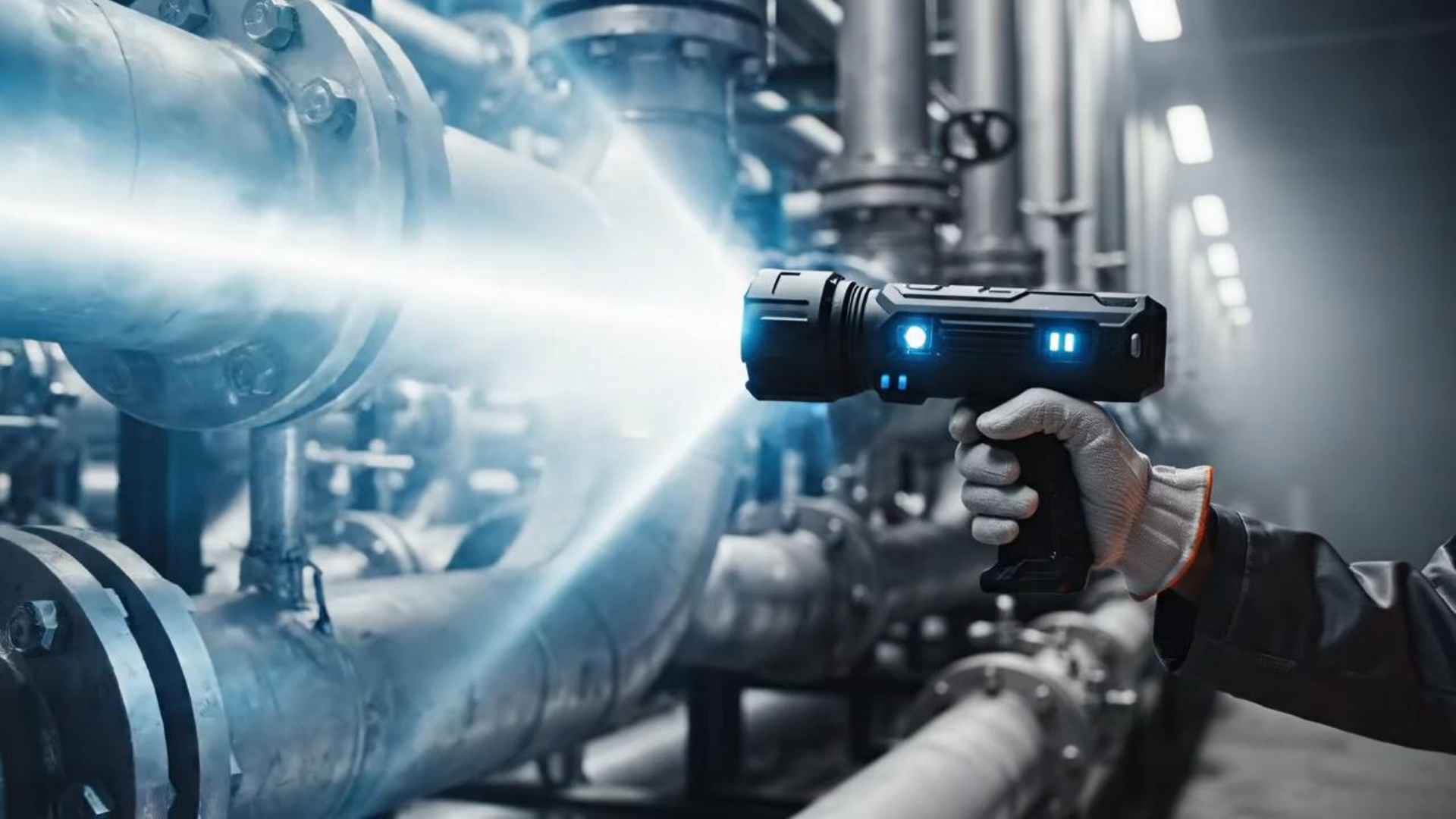
Thermal image projected onto the wall reveals structural hotspots and victim locations on the other side.

03

Shared Team View

Entire team sees the same projection—superior shared awareness vs. individual thermal goggles.





A Universe of Applications

The core technology is a platform. The primary sensor is modular; software is extensible via API.



Architecture

Project blueprints and structural integrity scans onto buildings in real scale.



Logistics

Scan a pallet and project contents, routing, and destination onto boxes.

Open API



Retail

Project customer reviews and live product information directly onto shelf space.



Education & Medical

Project anatomical diagrams onto training manikins for immersive learning.

Market Opportunity & Competitive Edge

\$70B+

AR Enterprise Market by 2027



Collaborative by Design

Unlike isolating headsets, LUMINAR creates a shared view for entire teams.



Low Friction

No wearables, no complex setup. Point and see — instant insight.



Cost-Effective

Lower BOM and simpler design than high-end AR glasses.



Safety & Durability

Built as a rugged tool for demanding environments.

Screen-Bound



Tablets



AR Headsets

World-Integrated



LUMINAR

The Core Team: Specialized Expertise Required



Project Lead / Manager

Guides vision, manages agile sprints, clears roadblocks.



Hardware & Embedded Systems

Designs PCB, sensor integration, and power management — architects of the physical product.



Computer Science Engineers

Builds OS, data pipelines, and critical projection mapping algorithms — the 'brains.'



Computer Vision & ML Engineers

Develops perception, object recognition, sensor fusion. Teaches the device to 'see.'



Graphics & UI/UX Engineers

Designs 'reinforeographics' ensuring projected data is intuitive and interactive.



QA & Test Engineers

Builds automated and manual testing frameworks for reliability, accuracy, and durability.

Software Core: Why CS is Our Bedrock

The Geometry Problem

CS / Graphics

Real-time projection mapping onto arbitrary, non-flat surfaces requires keystone correction, distortion mapping, and focus adjustment — a pure computational geometry challenge.

The Speed Problem

CS / Embedded

Processing massive LiDAR point-clouds and high-framerate thermal video with sub-second latency demands deep knowledge of data structures, memory management, and C++/GPU programming.

The Integration Problem

CS / Systems

Building stable, low-level firmware and OS that allows sensors, processor, and projector to communicate without bottlenecks is fundamental systems engineering.

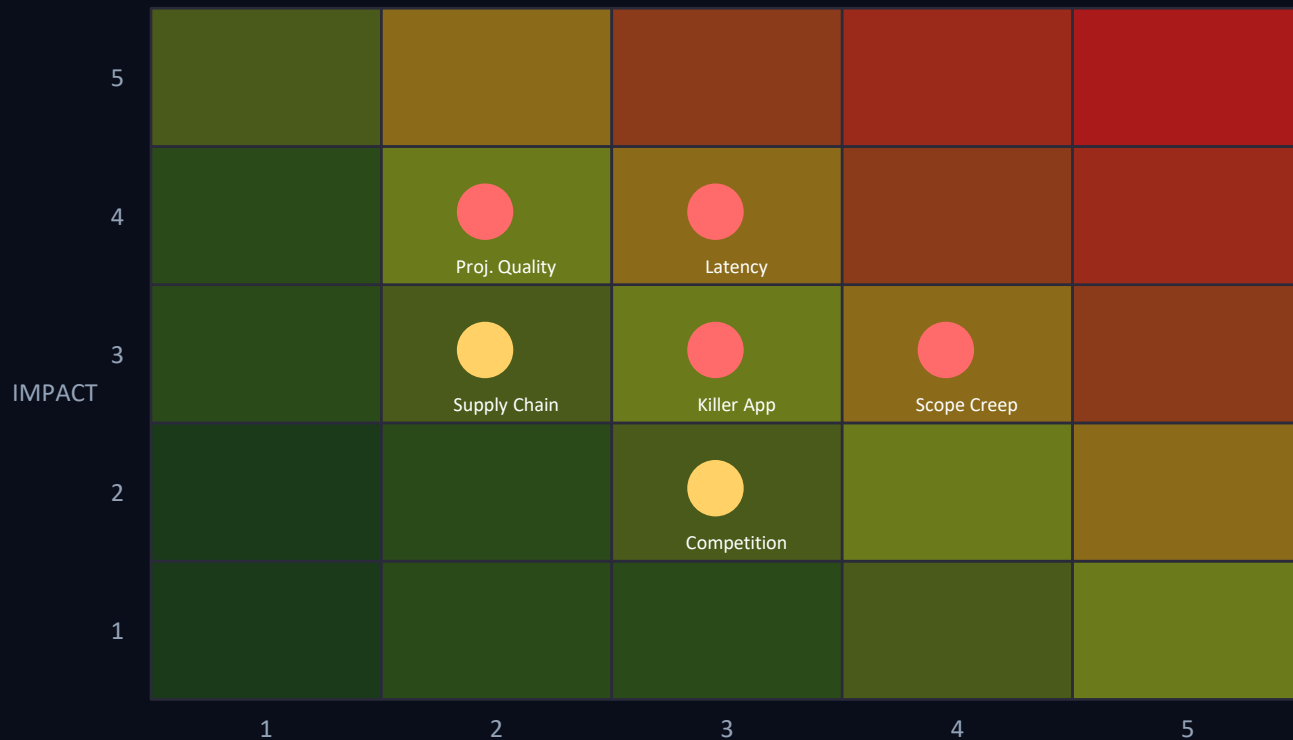
The Scalability Problem

CS / API Design

A robust API is essential for our platform vision — enabling future developers to create 'reinfographic apps' for new sensors and use cases.

Proactive Risk Management Framework

We assess risks based on Likelihood and Impact to prioritize mitigation efforts.



RISK KEY

- Latency
- Proj. Quality
- Sensor Fusion
- Battery Life
- Competition
- Killer App
- Scope Creep
- BOM Cost
- Supply Chain

Technical & Engineering Risks

R1

Projection Quality in Ambient Light

Impact: High

Mitigation: Source state-of-the-art laser-based pico-projectors. Implement ambient light sensor to auto-adjust contrast and brightness in software.

R2

Real-Time Processing Latency

Impact: High

Mitigation: Utilize a powerful SoC with dedicated NPUs. CS team focuses on algorithm optimization using C++ and GPU shaders. Benchmark and profile relentlessly from day one.

R3

Sensor Fusion & Calibration

Impact: Medium

Mitigation: Develop rigorous, automated factory-level calibration for each device. CV/ML engineers implement advanced sensor fusion algorithms (e.g., Extended Kalman Filters) to maintain alignment in real-time.

R4

Battery Life

Impact: Medium

Mitigation: Design a robust power management system with user-swappable battery pack. Select energy-efficient components and implement multiple software power-saving modes.

Market & Project Management Risks

R5

Competition from AR Headsets

Impact: Medium

Mitigation: Aggressively market key differentiators: collaboration, ease of use, and lower cost. Focus on use cases where headsets are impractical (shared-space training, quick inspections).

R6

Lack of a 'Killer App' at Launch

Impact: High

Mitigation: Laser-focus initial MVP development on the thermal imaging application for industrial/emergency services. Secure a pilot partner to co-develop features and ensure product-market fit before expanding.

R7

Scope Creep

Impact: High

Mitigation: Strict adherence to Agile methodology with a well-defined MVP. Project Manager empowered to ruthlessly defend the roadmap and push non-essential features to V2 backlog.

Financial & Supply Chain Risks

R8

High Bill of Materials (BOM) Cost

Impact: Medium

Mitigation: Engage in extensive component sourcing and negotiation early. Design for Manufacturing and Assembly (DFMA) from the start. Explore strategic partnerships with key component manufacturers.

R9

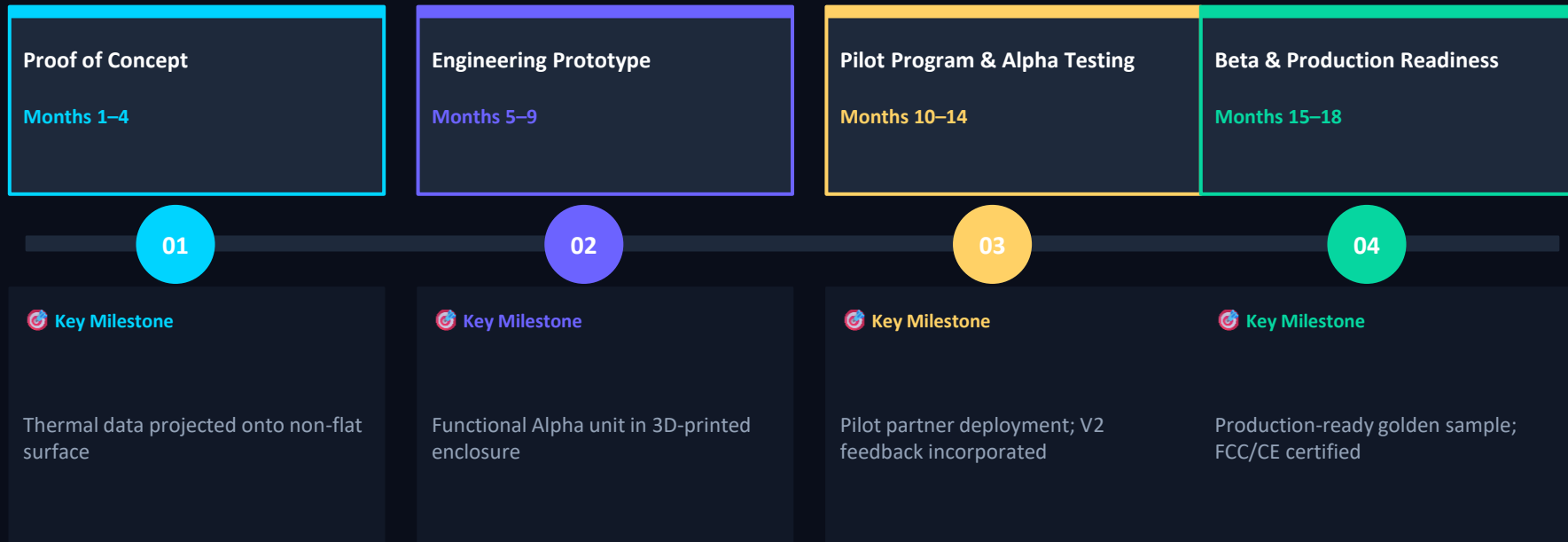
Supply Chain Disruption

Impact: High

Mitigation: Proactively identify and qualify second-source suppliers for all critical components (SoC, LiDAR, Projector). Maintain strategic buffer inventory and build relationships with multiple distributors.

Project Roadmap & Timeline

A clear, four-phase plan from concept to market — 18 months total.



Detailed Engineering Process by Phase

P1 Proof of Concept

- Objectives:** Validate core algorithms, test sensor accuracy.
- Tasks:** Develop projection mapping code; build sensor fusion model; test projector brightness.
- Focus:** CS & CV/ML Engineers (algorithm dev), Hardware Engineer (component integration on dev kits).

P2 Engineering Prototype

- Objectives:** Custom hardware design, firmware development, industrial design.
- Tasks:** PCB layout; write low-level drivers; design ergonomic enclosure; power management system.
- Focus:** Hardware & Embedded Systems (primary), CS Engineers (porting PoC code to custom hardware).

P3 Pilot Program

- Objectives:** Stability, reliability, and user feedback integration.
- Tasks:** Bug fixing; UI/UX refinement from field tests; build automated test suites.
- Focus:** QA & Test Engineers (primary), UI/UX & Graphics Engineers, CS Engineers (feature refinement).

P4 Production Readiness

- Objectives:** Design for Manufacturing (DFM), certification, final OS hardening.
- Tasks:** Finalize BOM; pass FCC/CE testing; optimize boot time and battery life.
- Focus:** Hardware Engineers (factory liaison), Embedded Systems Engineers (final firmware).

Estimated Budget & Resource Allocation

Seed Funding
Target

\$750K

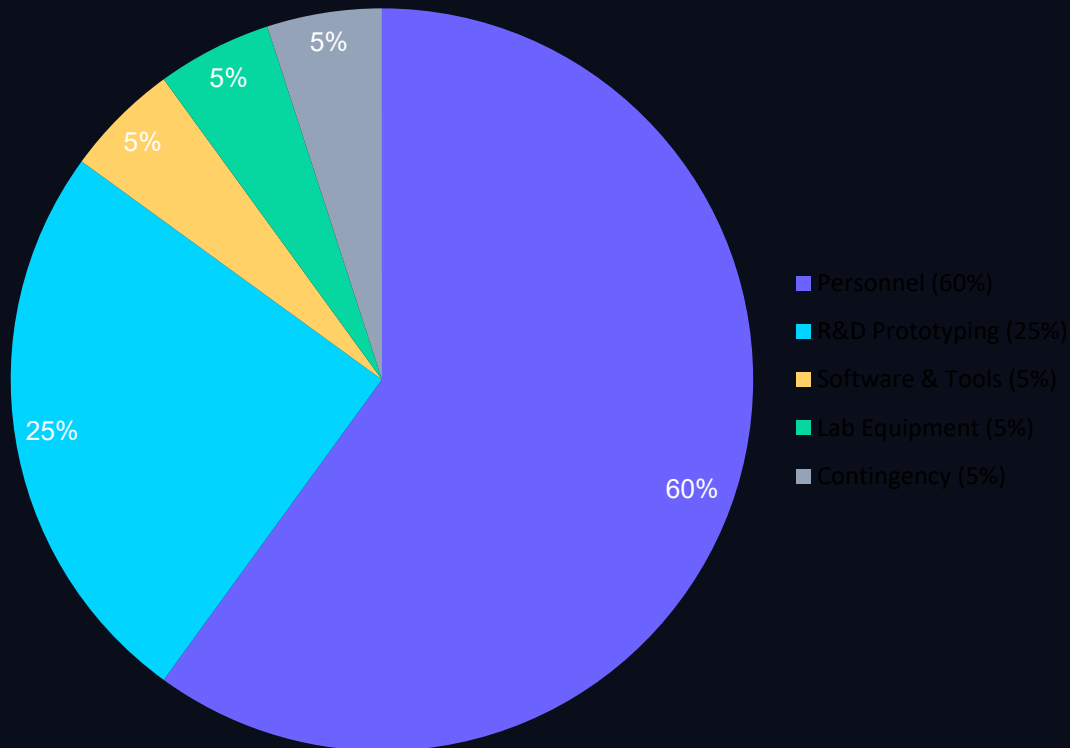
Phases 1 & 2 — Proof of Concept through
Engineering Prototype



Personnel (60%): Salaries for CS, Hardware, CV/ML engineers — our greatest asset.



R&D Prototyping (25%): LiDAR sensors, projectors, dev kits, PCB fabrication, 3D printing.



The Ask & Next Steps

Seeking \$750,000 in Seed Funding

to complete Phases 1 & 2 and deliver a functional handheld Alpha prototype

1

Hire Core Team

Recruit the 4-person engineering team (CS, Hardware, CV/ML, Embedded Systems).

2

Procure & Build

Acquire all necessary components and equipment for prototyping.

3

Alpha in 9 Months

Deliver a functional, handheld Alpha prototype ahead of Phase 3 pilot program.

⚡ **Immediate Next Step:** Secure funding to begin hiring our lead Computer Science and Hardware Engineers.

P R O J E C T

LUMINAR

The Future is Projected

LUMINAR is more than a device — it's a new platform for interacting with data.

Intuitive, collaborative, and built for the real world.

We have the vision, the plan, and a clear understanding of the challenges.

Join us in liberating data from the screen.

Q&A

Let's discuss how we can build this future together.

PROJECT LUMINAR · From Vision to Viability

Confidential — Seed Funding Presentation